

SQL Aggregate Functions with NULL

Easy Notes & Exam Special — Behavior of NULL in Aggregate Calculations

1. What is NULL?

NULL does **NOT** mean 0.

- 0 → An actual, known numeric value.
- NULL → Value is missing, unrecorded, or unavailable.

SAMPLE DATASET: TABLE T1 (5 ROWS)

Row #	Column A	Column B
Row 1	1	NULL
Row 2	2	NULL
Row 3	3	NULL
Row 4	4	NULL
Row 5	NULL	NULL

2. BEHAVIOR OF COUNT() WITH NULL

COUNT(*)

Output: 5

Counts total rows in the table. Row 5 (NULL | NULL) is still counted!

```
SELECT COUNT(*) FROM T1;
```

COUNT(A)

Output: 4

Column A contains 1, 2, 3, 4, NULL. Ignores the 1 NULL value.

```
SELECT COUNT(A) FROM T1;
```

COUNT(B)

Output: 0

Column B contains only NULL values across all 5 rows. Result is 0.

```
SELECT COUNT(B) FROM T1;
```

★ Summary of COUNT Differences:

COUNT(*)	5	Counts every physical row
COUNT(A)	4	Counts non-NULL values in A
COUNT(B)	0	No non-NULL values exist in B

3. MATHEMATICAL AGGREGATES ON COLUMN A

SUM(A)

Output: 10

Calculation: `1 + 2 + 3 + 4 = 10` (`NULL` is ignored).

`SELECT SUM(A) FROM T1;`

AVG(A)

Output: 2.5

Calculation: `SUM(A) / COUNT(A)` → `10 / 4 = 2.5` (Divides by 4, NOT 5!).

`SELECT AVG(A) FROM T1;`

MIN(A) & MAX(A)

MIN: 1

MAX: 4

Evaluates available non-NULL set `{1, 2, 3, 4}`. Ignores `NULL`.

`SELECT MIN(A), MAX(A) FROM T1;`

4. WHAT HAPPENS WHEN ALL VALUES ARE NULL? (COLUMN B)

When an aggregate function is executed on a column containing **only NULLs** (like Column B):

SUM(B), AVG(B), MIN(B), MAX(B)

Output: NULL

Because no non-NULL values exist to compute, all mathematical aggregations return `NULL`.

`SELECT SUM(B), AVG(B), MIN(B), MAX(B) FROM T1;`

🔥 COMPREHENSIVE COMPARISON MATRIX

Aggregate Function	Column A (Mixed)	Column B (All NULL)
<code>COUNT(*)</code>	5	5
<code>COUNT(column)</code>	4	0
<code>SUM(column)</code>	10	NULL
<code>AVG(column)</code>	2.5	NULL
<code>MIN(column)</code>	1	NULL
<code>MAX(column)</code>	4	NULL

🧠 Golden Rules for Exams & Interviews:

1. `COUNT(*)` counts rows, including rows containing `NULL`.

2. `COUNT(column)` counts only non-NULL entries in that specific column.

3. For `SUM`, `AVG`, `MIN`, and `MAX`, `NULL` values are ignored when computing results.

4. If **all values** in a column are `NULL`, `COUNT(column)` returns `0`, while `SUM`, `AVG`, `MIN`, and `MAX` return `NULL`.